Assignment - 1

Module 1 – Overview of IT Industry

1. What is a Program?

⇨ Program is a set of instructions for a computer to execute a specific tasks.

2. Write a simple "Hello World" program in two different programming languages of your choice. Compare the structure and syntax ? - LAB EXERCISE

⇨ Input :-

hello.c

```c
//including the header files
#include <stdio.h>

main(){
    printf("Hello World!");
}
```

Output:-

E:\C# Practices\TOPS LAB PRA

```
Hello World!
--------------------------------
Process exited after 0.05553 seconds with return value 0
Press any key to continue . . .
```

Compare Structure and Syntax between C language and python programming language

C Language	Python Language
Compiled Programming Language	Interpreted Programming Language
High Speed	Low Speed
Complex Syntax	Simple Syntax
Limited set of Built-in Functions	Extensive library of Built-in Functions
Manual Memory Management	Automatic Memory Management

3. Explain in your own words what a program is and how it functions.

⇨ Program is a set of instructions for a computer to execute a specific tasks.

How Program Functions Work :-

- **Written in Code :-** A Programmer writes the instructions using specific programming languages such as python, c etc..
- This source code is designed to be clear and logical.
- **Compilation or Interpretation:-** Source code is converted into machine code through the process called compilation , or read and executed line-by-line by an interpreter.
- **Execution:-** The computer's CPU carries out instructions sequentially, from start to finish.
- This might involve processing input, performing calculations, making decisions based on conditions, managing data and producing output.
- **Interaction:-** The program interacts with computer's hardware, such as memory(RAM) Storage drives, screen, keyboard and network connections to complete it's task and produce to desired result or user experience.

4. What is Programming?

⇨ Programming is the process of writing step-by-step instructions called code in a specific language that tells computer how to perform task or solve the problem.

5. What are the key steps involved in the programming process?

⇨ Here's some key steps of programming process:-

Step 1 : Analysis/Problem Definition:-

- Understand the problem, user needs, goals, inputs, and expected outputs.

Step 2 : Design :-

- Develop a solution strategy (algorithm) using pseudocode or flowcharts.

Step 3 : Coding/Implementation :-

- Translate the design into actual code using a chosen programming language, adhering to syntax.

Step 4 : Testing & Debugging :-

- Run the program to find and fix errors (bugs).

Step 5 : Documentation :-

- Create comments, user manuals, and technical guides for clarity and future maintenance.

Step 6 : Deployment & Maintenance :-

- Release the program to users.
- Ongoing updates, bug fixes, and feature additions.

6. Types of Programming Languages?

⇨ There are basically five types of programming languages.

- Procedural- oriented Programming Language (such as C, C++, Java)
- Object-oriented Programming Language (Such as Java, Python, PHP)
- Logical Programming Language (such as prolog)
- Functional Programming Language (such as Scala, Erlang)
- Scripting Language (PHP, Ruby)

7. What are the main differences between high-level and low-level programming languages?

⇨ Here's the main differences between high-level and low-level programming languages

Low-Level Languages	High-Level Languages
Easy to Understand for machine, but less human-readable.	Human readable and easier to understand and use.
Not Portable	Portable
Higher Execution Speed	Comparatively Lower Execution Speed
Uses Binary code and mnemonics	Uses Syntax and semantics
Example:- Assembly Language, Machine Language	Example:- Java, Python, C++

8. World Wide Web & How Internet Works ?

⇨ It is a collection of websites or web pages stored in web servers.

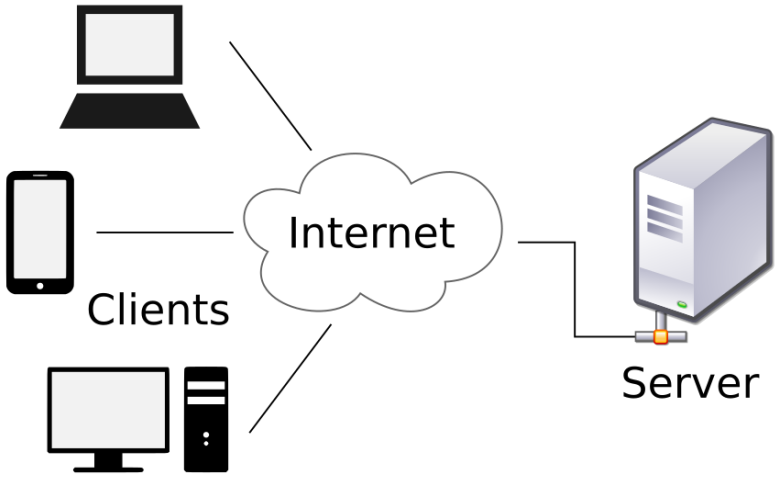
⇨ Connected to local computers through the internet.

How Internet Works :-

- The Internet works by connecting networks together through a series of routers and switches.

9. Research and create a diagram of how data is transmitted from a client to a server over the internet. -- LAB EXERCISE

⇨



10. Describe the roles of the client and server in web communication.

- ⇨ **Client sent a Request :-** The client start the conversation by a sending a request.
- ⇨ **Server Processes Request :-** The server does necessary work – this might involve querying a database, running a computation, or reading a file.
- ⇨ Server Sends a Response :- The server sends back the requested information, or an error message if something goes wrong.

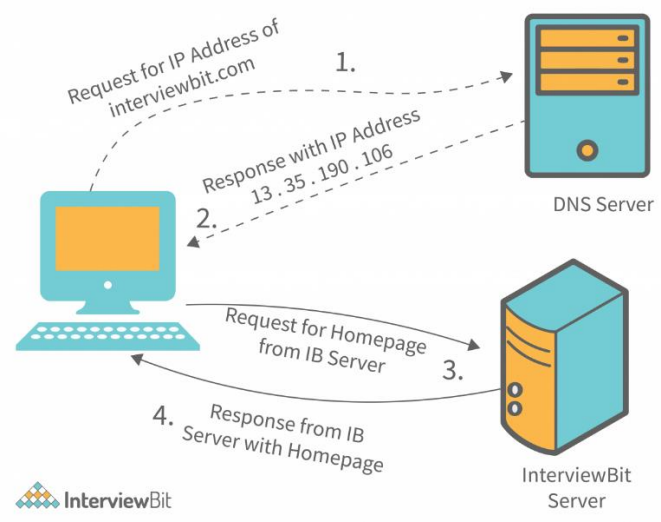
11. Network Layers on Client and Server ?

⇨ There are basically 7 layers on Client and Server.

- ➢ Physical Layer :- Deals with the physical connection between devices.
- ➢ Data-Link Layer :- Handle communication between adjacent network nodes.
- ➢ Network Layer :- Manages data transfer between different networks.
- ➢ Transport Layer :- Ensures reliable transmission of data.
- ➢ Session Layer :- Manages sessions and connections between applications.
- ➢ Presentation Layer :- Translate the data between application layer and the network.
- ➢ Application Layer :- Interfaces directly with the end-user or application.

12. Design a simple HTTP client-server communication in any language. - LAB EXERCISE:

⇨



13. Explain the function of the TCP/IP model and its layers ?

- ⇨ TCP/IP Working :- Breaks data into segments and ensures reliable (TCP) or Fast (UDP) delivery.
- ⇨ Layers of TCP/IP Model :-

    - ➢ Application Layer :- This layer is used by users and application, it include protocols like HTTP, FTP, SMTP. It manages how the data is presented, formatted and secured.
    - ➢ Transport Layer :- Ensures reliable transmission of data.
    - ➢ Network Layer :- Manages data transfer between different networks.
    - ➢ Network Access Layer:- Handles how data is sent on the local network using hardware, cables, Wi-Fi, Ethernet, etc.

14. Explain Client Server Communication ?

- ⇨ **Client sent a Request :-** The client start the conversation by a sending a request.
- ⇨ **Server Processes Request :-** The server does necessary work – this might involve querying a database, running a computation, or reading a file.
- ⇨ **Server Sends a Response :-** The server sends back the requested information, or an error message if something goes wrong.

15. Types of Internet Connections ?

- ⇨ There are 8 types of Internet Connections:

    - ➢ Dial-Up Connection
    - ➢ Broadband Connection
    - ➢ DSL(Digital Subscriber Line)

- Cabel
- Satellite Connection
- Wireless Connection
- Cellular
- ISDN (Integrated Service Digital Network).

16. Research different types of internet connections (e.g., broadband, fiber, satellite) and list their pros and cons. - LAB EXERCISE

⇨ Here's are the different types of internet connections

- Dial-Up Connection :-

  Pros :- Low Cost.

  Cons :- Very Slow speed.

- Broadband Connection :-

  Pros :- High-Speed Internet, Always Connected.

  Cons :- More expensive than dial-up connection.

- DSL(Digital Subscriber Line) :-

  Pros :- Uses existing telephone lines, Faster than Dial-Up

  Cons :- Speed decreases with distance

- Cabel :-

  Pros :- Fast Internet Speed, Goof for Streaming.

  Cons :- Speed may slow during peak hours.

- Satellite Connection :-

  Pros :- Available in Remote Areas.

  Cons :- Affected by weather.

- Wireless Connection :-

  Pros :- No Cable Required, Easy to use.

Cons :- Limited Range

- Cellular Connection :-

  Pros :- Portable and mobile access

  Cons :- Limited data Plans.

- ISDN(Integrated Service Digital Network) :-

  Pros :- More reliable than dial-up.

  Cons :- Outdated and expensive.

17. How does broadband differ from fiber-optic internet?

⇨ Here's a difference from broadband and fiber-optic

Broadband Internet	Fiber-Optic Internet
Medium to High Speed	Very High Speed
Uses Copper Cables (DSL/Cable)	Uses Fiber-Optic Cables
Can slow during peak hours	Very reliable
Widely available	Limited availability
Less Expensive	More Expensive
Best for home use	Best for gaming, streaming, heavy usages.

18. Protocols ?

⇨ Protocol is a set of rules accomplished by network.
⇨ Protocols identifies the rules, syntax, semantics and synchronization of communication and error managing methods.
⇨ There are five types of protocols

- HTTP or HTTPS
- FTP (File Transfer Protocol)
- Email (POP3, SMTP)
- TCP (Transmission Control Protocol)
- UDP (User Datagram Protocol)

19. Simulate HTTP and FTP requests using command line tools (e.g., curl) - LAB EXERCISE ?

- HTTP Request using curl :-

  Curl https : //google.com

- Sends an http GET request and show the web page content.

- HTTP POST Request :-

  Curl – X POST – d “name=yash” https://google.com/form

- Sends data to a server.

- FTP Request using curl :-

  Curl ftp://ftp.google.com

- Lists files on a FTP Server.

- FTP Download :-

  curl -u username : password ftp://ftp.google.com/file.txt

20. What are the differences between HTTP and HTTPS protocols?

- Here’s the difference of HTTP and HTTPS Protocols

HTTP	HTTPs
HTTP stands for Hyper Text Transfer Protocol.	HTTPs stands for Hyper Text Transfer Protocol Secure.
HTTP is less secure.	HTTPs is more secure.
HTTP works in application layer of OSI model.	HTTPs works in transport layer of OSI model.
HTTP does not use certificate.	HTTPs uses SSL certificate.
There is no encryption and decryption.	There is encryption and decryption.

21. What is Application Security ?

- It refers to security precautions used at the application level to prevent the theft to hijacking of data or code within the Application.

- ⇨ It include security concerns made during application development and design etc. once they have been deployed.

22. Identify and explain three common application security vulnerabilities. Suggest possible solutions. - LAB EXERCISE

- ⇨ Here's a three common application security vulnerabilities :-

- ➢ SQL Injection :-

- ⇨ Attackers insert malicious SQL code into input fields to access or modify database.

Solutions :-

- ⇨ Use Prepared Statement / Parameterized queries.
- ⇨ Validate and sanitize user input.
- ⇨ Use ORM Frameworks.

- ➢ Cross-Site Scripting (XSS) :-

- ⇨ Attackers inject malicious scripts into web pages viewed by other users.

Solutions :-

- ⇨ Escape and encode output.
- ⇨ Use Content Security Policy (CSP).
- ⇨ Validate and sanitize user input.

- ➢ Broken Authentication :-

- ⇨ Weak authentication allows attackers to take over user accounts.

Solutions :-

- ⇨ Enforce strong passwords policy.
- ⇨ Use Multi-Factor Authentication (MFA).
- ⇨ Secure Session Management (HTTPs, secure cookies).

23. What is the role of encryption in securing applications?

- ⇨ Encryption ensures no one can read communications or data except the intended recipient or data owner.
- ⇨ This prevents attackers from intercepting and accessing sensitive data.

24. Software Applications and Its Types ?

- ⇨ Software applications are programs designs to help users perform specific tasks on a computer or mobile device.
- ⇨ Types of Software Applications :-

- ➢ Application Software
- ➢ System Software
- ➢ Driver Software
- ➢ Middleware
- ➢ Programming Software

25. Identify and classify 5 applications you use daily as either system software or application software ? - LAB EXERCISE

- ⇨ Here's the five applications I daily use either system software or application software.

- ➢ Google Chrome (Web Browser) :- Classification :- Application Software
- ➢ WhatsApp (Messaging) :- Classification :- Application Software
- ➢ Spotify (music streaming) :- Classification :- Application Software
- ➢ Google Maps (Navigation) :- Classification :- Application Software
- ➢ Android/IOS (Operating System) :- Classification :- System Software.

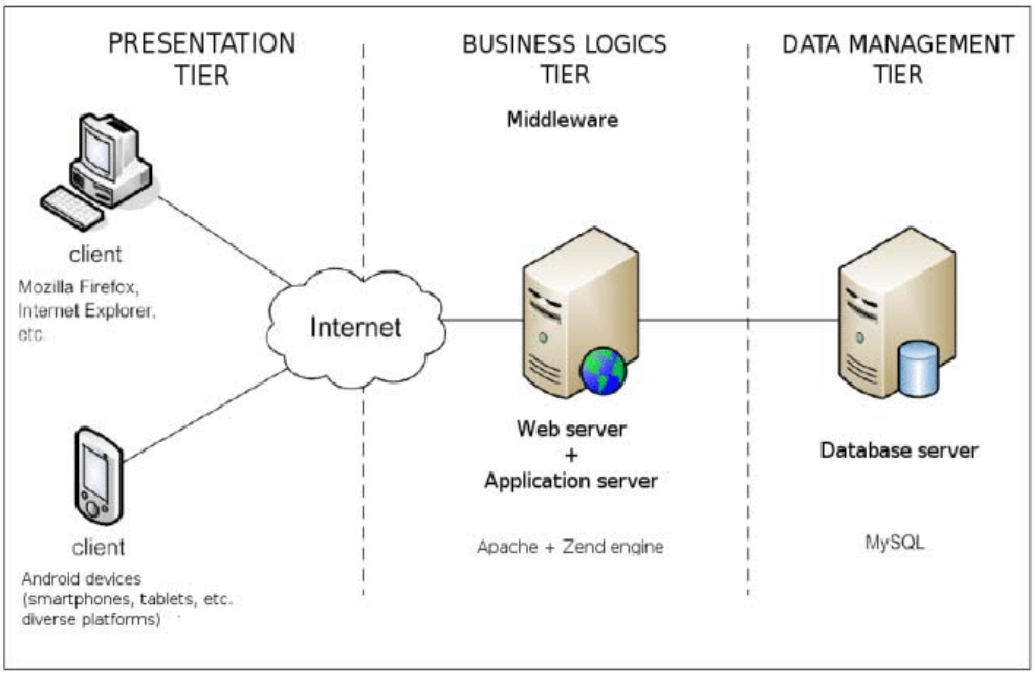
26. What is the difference between system software and application software?

- ⇨ Here's the difference between system software and application software.

System Software	Application Software
Manages and controls Computer Hardware.	Helps user perform specific tasks.
Acts as an interface between hardware and user.	Acts as an interface between user and system software.
Runs in the background.	Runs when user starts it.
Required for system to operate.	Not Required for basic system operation.
Installed during system setup.	Installed as per user needs.
Examples :- Operating Systems, Device Drivers.	Examples :- MS Word, Excel, Browser.

27. Design a basic three-tier software architecture diagram for a web application. – LAB EXERCISE

⇨ Here's below the diagram



28. What is the significance of modularity in software architecture?

⇨ Here's the below significance of modularity

➢ **Improves Maintainability :-**

- Makes the system easier to understand.
- Allow changes and bug fixes without affecting other parts.

➢ **Enhances Reusability :-**

- Modules can be reused in different projects.
- Saves development time and efforts.

➢ **Supports Scalability :-**

- Individual modules can be optimized, replaced or scaled independently.
- New functionality can be added without redesigning the entire system.

➢ **Enables Parallel Development :-**

- Different teams can work separate modules at the same time.
- Reduce development time.

➢ **Improves Testability :-**

- Unit Testing of individual modules.
- Make it easier to identify and fix errors.

➢ **Promotes Flexibility and Adaptability :-**

- Modules can be updated or swapped with minimal impact
- The systems adapts more easily to new technologies or platforms.

➢ **Encourages Better design practice :-**

- Support separation of concerns.
- Ensure low coupling and high cohesion.

➢ **Reduces Risk and Cost :-**

- Limits the impacts of changes.
- Lowers long-term maintenance costs.

29. Layers in Software Architecture ?

⇨ There are basically five types of layer in software architecture:-

➢ Presentation Layer :-

- It is also called UI layer.
- It handles everything related to the user interface , displaying data to users and collecting input from them.
- This tire is most accessible

➢ Application Layer :-

- It handles main programs of architecture.
- It include code definitions and most basic functionality.
- This is the layer that programs spend most of the time when working on software.

➢ Business Layer :-

- It is also called domain layer.

- This layer is contain a business logic – the core rules defines how the system behaves and process the data.

➢ Persistence Layer :-

- It is also called data-access layer.
- It manages data access, such as querying databases , mapping objects to relational data (ORM).

➢ Database Layer :-

- It stores and retrieve the application data, typically in database like SQL or NoSQL.

30. Create a case study on the functionality of the presentation, business logic, and data access layers of a given software system ? - LAB EXERCISE

⇨ Here's the layers functionality :-

➢ Presentation Layer :-

- It is also called UI layer.
- It handles everything related to the user interface , displaying data to users and collecting input from them.
- This tire is most accessible

➢ Business Layer :-

- It is also called domain layer.
- This layer is contain a business logic – the core rules defines how the system behaves and process the data.

➢ Persistence Layer :-

- It is also called data-access layer.
- It manages data access, such as querying databases , mapping objects to relational data (ORM).

31. Why are layers important in software architecture?

⇨ There are several key reasons that are layers are very important in software architecture :-

- **Separation of Concerns**

- **Improved Maintainability**
- **Better Reusability**
- **Enhanced Scalability**
- **Improved Testability**
- **Flexibility and Adaptability**
- **Supports Team Collaboration**
- **Improves Security**

➢ this above reasons that's layer are very important.

32. Explore different types of software environments (development, testing, production). Set up a basic environment in a virtual machine. - LAB EXERCISE

⇨ Here's a below types of software environments :-

➢ Development Environment :-

- Where Developers write and modify the code. - **purpose**
- Frequent code changes.
- Debugging data enabled.
- Uses test/dummy data.

➢ Testing Environment :-

- To verify that software works correctly before release. - **purpose**
- More stable than Development Environment.
- Uses test data.

➢ Production Environment :-

- The Live system used by real users. – **purpose**
- Very stable and secure.
- Uses Real data.

⇨ **Setup a basic environment of Virtual Machine :-**

Here's a below steps :-

Step 1 :- Install Virtualization Software

- Choose One :- Virtual Box or VMware Workstation Player.

Step 2 :- Download Linux ISO

- Ubuntu 22.04 LTS --- Recommended

Step 3 :- Create a Virtual Machine

- Open Virtual Box
- Click New
- Name : Dev-Environment
- Type : Linux
- Version : Ubuntu (64 bit)
- Memory : 2-4 GB RAM
- CPU : 2 cores
- Disk : 20-30 GB (VDI, dynamically allocated)

Step 4 :- Install Ubuntu in VM(Virtual Machine)

- Attach the ubuntu ISO file to the VM
- Start VM
- Choose Install Ubuntu
- Follow on-screen instructions
- Create a username and password
- Complete Installation and then Reboot the system.

Step 5 :- Update the System

- **Open terminal inside the VM and run :**

  sudo apt update && sudo apt upgrade -y

Step 6 :- Install basic development tools

- sudo apt install -y git curl build-essential.

33. Explain the importance of a development environment in software production.

⇨ Here's below importance of development environment in software production

➢ **Safe experimentation :-** Developers can write and test the code without affecting the live server.

- **Early bug detection :-** Errors are found and fixed it early, reducing cost issue later.
- **Faster Development :-** Tools like IDs, Debuggers and libraries speed up coding.
- **Consistency:-** Ensures all developers work with the same tools and configuration.
- **Better Quality Software :-** Continuous testing and improvements lead to more stable and reliable products.

34. What is the difference between source code and machine code?

⇨ Here's the basic difference of source code and machine code

Source code	Machine code
Human Readable instructions written by programmers.	Binary Instructions directly understood by the computer.
High Level Language	Low Level Language
Easy for humans to read and understand	Very difficult for humans to read
It represents uses words, symbol and syntax.	It represents Binary digits
Easier to detect and fix errors	Errors are hard to detect and debug.
Example :- print("Hello World")	Example :- 10101010 00001010

35. Why is version control important in software development?

⇨ Here's are some reasons when version control are very important

- **Track Changes :-** keeps a full record of who change the code , what was changed and when?
- **Enables team collaboration :**- Allow many developers to work on the same project at the same time without conflicts.
- **Prevents data loss :**- Old version of file can be restored if something is deleted or goes wrong.
- **Support experimentation :**- Developers can test new features separately without affecting the main code.
- **Improves code quality :**- code can be reviewed , compared and tested before final approval.
- **Helps find and fixed bugs :**- Makes it easier to see when and where an error was introduced.
- **Maintains project history :**- store the complete development history for reference and learning.
- **Support CI/CD :**- Works well with automated testing and deployment systems.

36. What are the benefits of using GitHub for students?

⇨ There are many benefits of using GitHub below :-

- Free access to professional version control tools (such as github copilot, jetbrains IDEs.).
- Easy collaborations on group projects.
- Track and manage code changes safely.
- Build a public portfolio for employers.
- Learn industry-standard workflow(git).
- Access to GitHub student Developer Pack.
- Practice code review and teamwork.
- Showcase and share project easily.

37. Create a list of software you use regularly and classify them into the following categories: system, application, and utility software. - LAB EXERCISE

⇨ Here's I daily use software and below the classification :-

- **System Software :-** Operating System (such as like Windows, Android).
- **Application Software :-**
  - **Web Browsers :-** Google chrome, Firefox, Microsoft Edge etc..
  - **Productivity :-** Microsoft Word, Excel, Google docs.
  - **Communication :-** Zoom , Google Meet, WhatsApp.
  - **Entertainment :-** Netflix, Spotify, Amazon Prime, Jio Hotstar etc..
- **Utility Software :-**
  - **Antivirus :-** Windows Defender, Quick heal, Avast etc.
  - **File Compression :-** WinZip, 7Zip.
  - **Disk Cleanup**
  - **Backup Software**

38. What are the differences between open-source and proprietary software?

⇨ Here's below a basic difference of open-source and proprietary software

Open-source software	Proprietary software
Developed Collectively	Owned by a specific company or individual.
Source code made by public.	Source code not made by public.
Users can access, modify the source code.	Access to software is restricted.

Example :- Linux Operating System, Mozilla Firefox Browser	Example :- Microsoft office , Adobe Photoshop

39. How does GIT improve collaboration in a software development team?

⇨ Here's below some key aspects to improve collaboration in software development team:-

- **Version Control :-** Track all changes, so nothing is lost.
- **Parallel Work :-** Multiple developers can work on same projects at once.
- **Branching :-** Create separate branches to develop features safely.
- **Merging :-** Combine work from different team members smoothly.
- **Change history :-** See who changed what and when.
- **Rollback Support :-** Easily revert to previous versions if something breaks.
- **Code review :-** Helps review and discuss changes before merging.

40. Write a report on the various types of application software and how they improve productivity. - LAB EXERCISE

⇨ There are several types of application software

- **Word Processing Software :- (ex :- MS Word)**
  - Used for writing and editing documents.
  - Save time with formatting and spell check.
  - **Productivity Improvement :-** Faster document creation and reduced errors.
- **Spreadsheet Software :- (ex :- MS Excel)**
  - Used for calculation and data analysis.
  - Generates charts and reports quickly.
  - **Productivity Improvement :-** Improve accuracy and speeds up data processing.
- **Presentation Software :- (ex:- MS Power point)**
  - Used to create slides and presentations.
  - Makes information clear and attractive.
  - **Productivity Improvement** :- Saves time and enhance communication of ideas.
- **Database Software :- (ex:- MS Access)**
  - Stores and manages large data.

- Allow Quick data search and updates.
- **Productivity Improvement :-** Improve data organization and retrieval speed.

➢ **Communication Software :- (ex:- Zoom, MS Teams)**

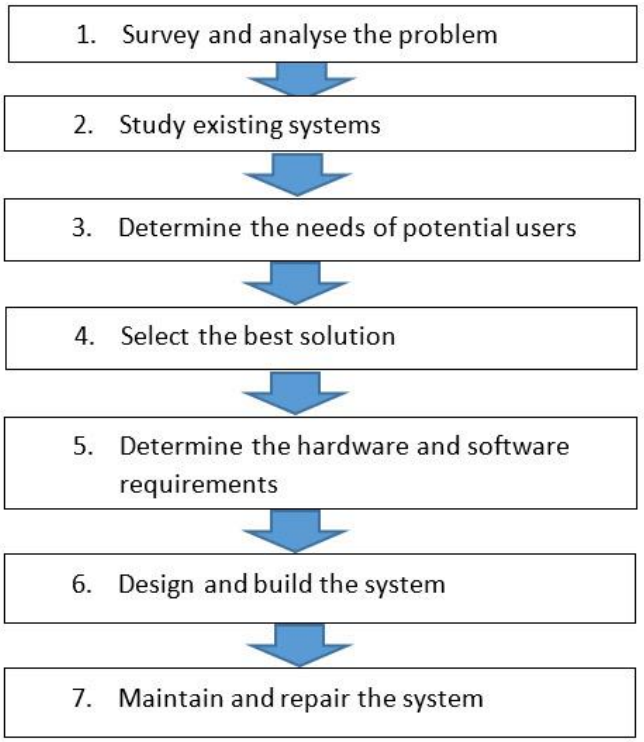
- Helps people communicates quickly.
- Supports remote work and virtual meetings.
- **Productivity Improvement :-** Reduced delays and improve teamwork**.**

41. What is the role of application software in businesses?

⇨ Application software helps businesses do everyday tasks like accounting, communication, data management and customer services, making work faster and more efficient.

42. Create a flowchart representing the Software Development Life Cycle (SDLC). - LAB EXERCISE

⇨ Here's the below flowchart that representing the software development life cycle



43. What are the main stages of the software development process?

⇨ There are basically 6 main stages of the software development process :-

- Requirement Gathering
- Analysis
- Designing
- Implementation
- Testing
- Maintenance

44. Write a requirement specification for a simple library management system. - LAB EXERCISE

⇨ Here's a requirement specification for a simple library management system.

- **Purpose :-** The system is design to help manage activities such as adding books, issuing books, returning books and keeping record efficiently.
- **Users :-**
  - Librarian(Admin)
  - Librarian members (Student/staff)
- **Functional Requirements :-**
  - The system shall allow the librarian to add, update and delete books record.
  - The system shall allow the librarian to register and manage members.
  - The system shall allow issuing and returning books.
  - The system shall track available and issued books.
  - The system shall generate basic reports.
- **Non-Functional Requirements :-**
  - The system shall easy to use.
  - The system shall responds quickly to user actions.
  - The system shall store data securely.
- **Constraints :-**

  - The system will be used on single computer or local network.
  - The system will use a simple database for storage.

45. Why is the requirement analysis phase critical in software development?

⇨ The requirement analysis phase is critical because it clearly defines what users needs, helps avoid misunderstandings, reduces errors and ensures the software built correctly form start.

46. Perform a functional analysis for an online shopping system. - LAB EXERCISE

⇨ Here's below functional analysis for an online shopping system

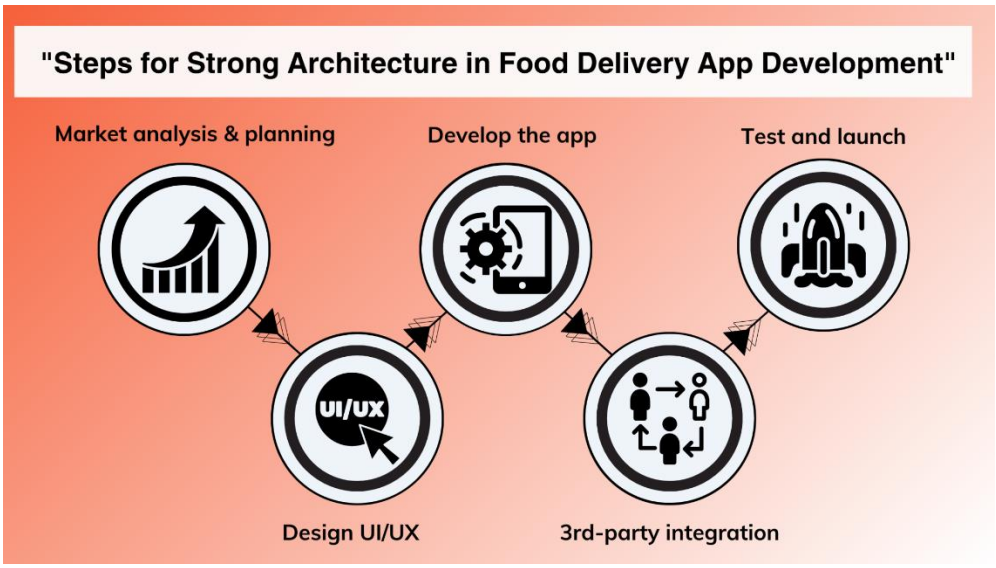
- User Registration and login
- Browse and search for products
- View product details and prices
- Add and remove item form shopping cart
- Place orders and make online payments
- Track order status
- Manage products and orders(Admin).

47. What is the role of software analysis in the development process?

⇨ The role of software analysis is to understand user needs, clearly define system requirements and ensure software is planned correctly before design and development begin.

48. Design a basic system architecture for a food delivery app. - LAB EXERCISE

⇨ Here's the basic system architecture for a food delivery app



49. What are the key elements of system design?

⇨ There are several key elements of system design

- Architecture
- Database Design
- APIs and Communication
- Caching
- Load Balancing

- Security
- Scalability and Performance
- Redundancy and Fault tolerance
- Monitoring and logging
- User Experience

50. Develop test cases for a simple calculator program. - LAB EXERCISE

⇨ Here's a simple calculator test cases scenario in table format

1. **Basic Arithmetic Operation Test cases :-**

TC ID	**Scenario**	**Input**	**Expected Output**
TC 01	Addition	10 + 5	15
TC 02	Subtraction	10 – 5	5
TC 03	Subtraction (Negative Result)	5 – 10	-5
TC 04	Multiplication	10 * 5	50
TC 05	Division	10 / 2	5
TC 06	Multiplication By Zero	10 * 0	0
TC 07	Decimal Calculation	3.5 + 2.5	6.0
TC 08	Negative Numbers	-5 + -3	-8

2. **Edge Case and Boundary Test Cases :-**

TC ID	**Scenario**	**Input**	**Expected Output**
TC 09	Division By Zero	5 / 0	Error (or cannot divided by zero)
TC 10	Large Number Calculation	999999999 + 1	Correct result and scientific notation
TC 11	Multiple Consecutive Operations	2 + 3 * 4	14 (Operator Precedence)
TC 12	Decimal Precision	0.1 + 0.2	0.3 (check floating point errors)
TC 13	Maximum/Minimum Input	Maximum digits allowed.	Proper display without truncation.

3. **Functional/Control Test Cases :-**

TC ID	Scenario	Input/Action	Expected Output
TC 14	Clear Function (C/AC)	5 + 5 'then press C'	Display resets to 0
TC 15	Backspace / Delete	Input 123, press backspace	Display shows 12
TC 16	Re-Calculation	5 + 5 = 10, then 2	2 (new input starts fresh)
TC 17	Equal Button Functionality	5 + 5	Pressing '=' results in 10

4. **Negative Test Cases :-**

TC ID	Scenario	Input/Action	Expected Output
TC 18	Invalid Input	Enter letters/Symbols	Error message or ignore input
TC 19	Empty Calculation	Press '=' without input	0 or error, no crash
TC 20	Invalid Operator Sequence	5 + + 5 or 5 + / * 5	Error or display first valid operator.

5. **UI/Usability Test Cases (For GUI Calculators) :-**

- Maximize/Minimize :- Ensure the window maximize and minimize properly.
- Button Responsiveness :- All digits and operators buttons responds to click.
- Display Formatting :- Large results are displayed without breaking the UI.

6. **Command-Line Specific Test Cases :-**

- Exit Command :- Ensures the application closes when a specific command is entered.
- Invalid menu selection :- If a menu(1,2,3,4) is shown, entering 5 should result in an invalid selection error.

51. Why is software testing important?

⇨ Here's a importance of software testing :-

1) Ensures Software works correctly
2) Finds Bugs early
3) Improves Product Quality
4) Enhances user satisfaction

5) Increases Security
6) Ensures reliability and performance
7) Reduces Business risks
8) Saves time and cost
9) Verify user requirements
10) Improves maintainability.

52. Document a real-world case where a software application required critical maintenance. - LAB EXERCISE

⇨ Real- world case :- Equifax Data Breach (2017)

- Organization :- Equifax (Credit Reporting Company)
- Software Used :- Apache Struts Web Application Framework

⇨ Critical Maintenance Issue :-

- Security Patch for Apache Struts was released in March 2017
- Equifax failed to update the affected application.
- The system continued running outdated, vulnerable software.

⇨ What Happened :-

- Hackers exploited the unpatched vulnerability.
- Unauthorized access occurred for several months.
- The breach was not detected immediately.

⇨ Impact :-

- Personal data of 147 million people was exposed.
- Included social security numbers and other sensitive data.
- Equifax faced huge financial losses legal penalties.

⇨ Lesson Learned :-

- Critical software updates must be applied on time.
- Regular Maintenance is essential for security and vulnerability.

53. What types of software maintenance are there?

⇨ There are basically four types of software maintenance

1) Corrective Maintenance :- To fix defects, error and bugs discovered after deployment.

2) Adaptive Maintenance :- To modify the software to remain compatible with changes in it's external environment
3) Perfective Maintenance :- To improve the software's performance, maintainability, and user-friendliness.
4) Preventative Maintenance :- To proactively prevent future problems, reduce complexity and improve system robustness.

54. What are the key differences between web and desktop applications?

⇨ Here's below the basic difference between web and desktop applications.

Desktop Application	Web Application
App runs locally	App runs in the cloud
Offline access	Internet access required
Security	Cyberattack risks
Manual updates	Automated updates
High speed	Speed depends on the lot of factors

55. What role does UI/UX design play in application development?

⇨ Here's below roles of UI/UX :-

- **Roles of UX(User Experience) :-**

  - ✓ Makes the app easy and intuitive to use.
  - ✓ Focuses on user needs through research.
  - ✓ Creates positive emotions and satisfaction.
  - ✓ Improve engagement and reduces user drop-off.

- **Roles of UI (User Interface) :-**

  - ✓ Focuses on visual looks (color, fonts, layout)
  - ✓ Create a first strong impression.
  - ✓ Maintain brand consistency and identity.

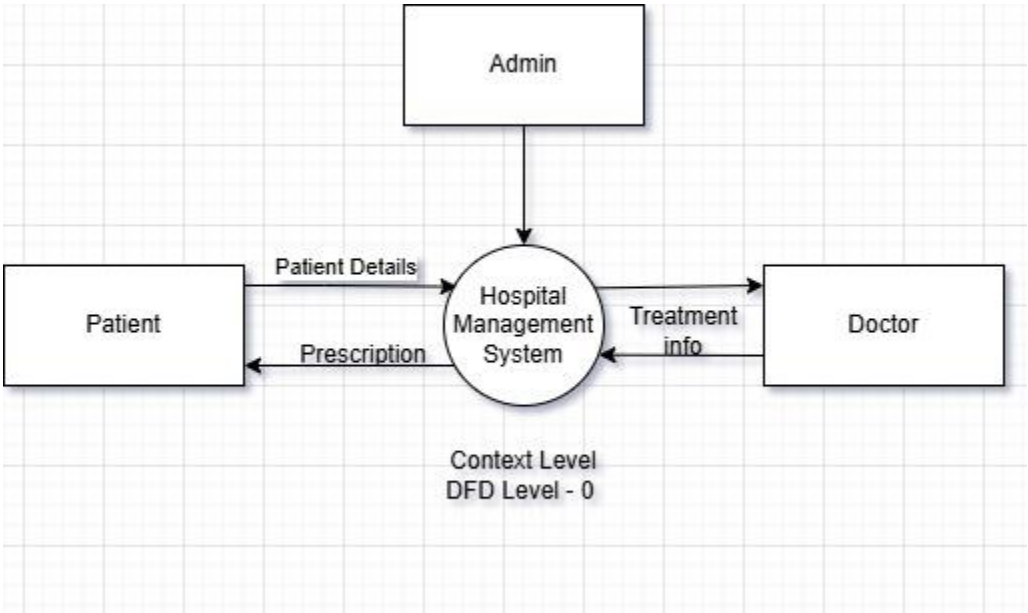
56. What are the differences between native and hybrid mobile apps?

⇨ Here's a below difference between native and hybrid mobiles apps

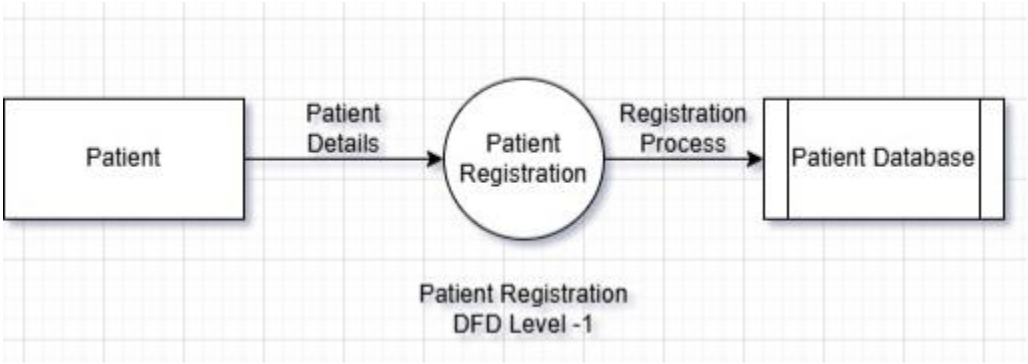
Native Apps	Hybrid Apps
Platform specific functionality	Multiple Functionality
Operating speed Fast	Operating speed Medium
High-End Graphics Available	High-End Graphics Not Available
UI/UX High	UI/UX Moderate
Access to API Full	Restricted
App store Availability Available	App store Availability Available

57. Create a DFD for a hospital management system. - LAB EXERCISE

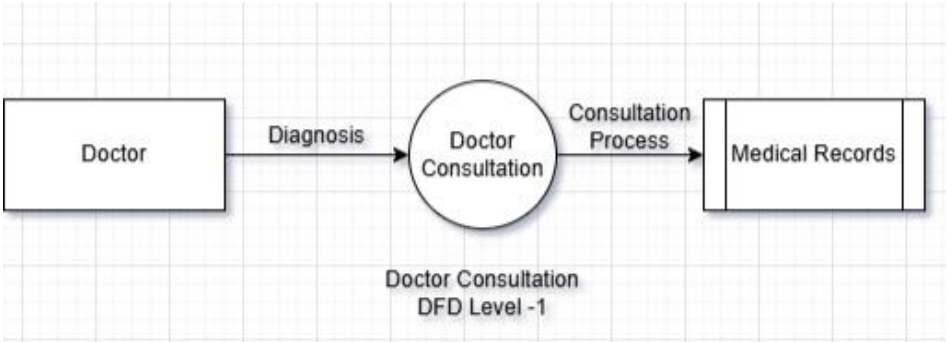
⇨ Below the DFD of a hospital management system:-



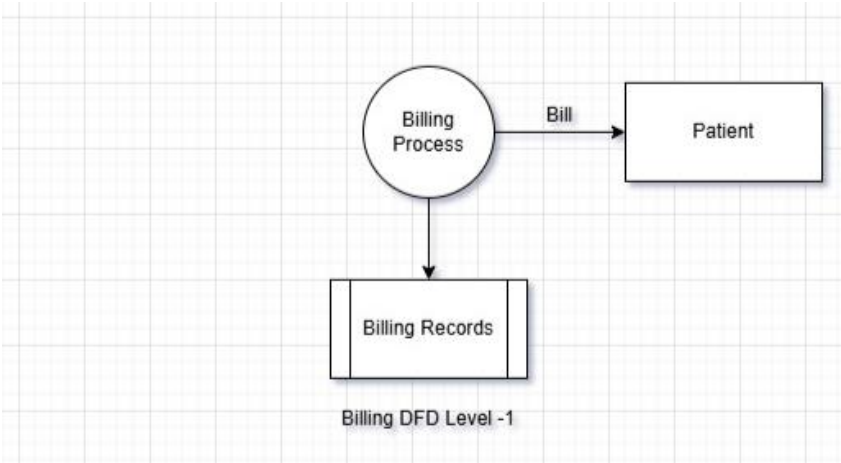
DFD Level – 0 of Hospital Management System



DFD Level – 1 Patient Registration



DFD Level – 1 Doctor Consultation



DFD Level – 1 Billing Generate

58. What is the significance of DFDs in system analysis?

⇨ Significance of DFDs :-

- ✓ Shows data flow clearly and simply.
- ✓ Helps understand how system works.
- ✓ Identifies errors, delays and inefficiencies.
- ✓ Improves communication among team members.
- ✓ Support system design and documentation.
- ✓ Breaks complex systems into smaller levels.
- ✓ Helps locate sensitive data for security.

59. Build a simple desktop calculator application using a GUI library. - LAB EXERCISE

⇨ Here's below code :-

⇨ **Input :-**

(globals)

calculator.c

```c
#include<stdio.h>
#include<float.h>

int main(){
    //declaring a varaible
    char op;
    double a,b,res;

    //print the output
    printf("Enter Operator (+,-,*,/): ");
    scanf("%c",&op);

    //read the two number
    printf("Enter the first number:");
    scanf("%lf",&a);

    printf("Enter the second number:");
    scanf("%lf",&b);

    //switch case
    switch(op){
        case '+':
            res = a + b;
        break;

        case '-':
            res = a - b;
        break;

        case '*':
            res = a * b;
        break;

        case '/':
            res = a / b;
        break;
        default:
            printf("Error! Invalid Operator");
            res = -DBL_MAX;
    }
    if(res!=-DBL_MAX)
        printf("%.2lf",res);

    return 0;
}
```

⇨ Output :-

E:\C# Practices\TOPS LAB PRA

```
Enter Operator (+,-,*,/): +
Enter the first number:10
Enter the second number:20
30.00
--------------------------------
Process exited after 7.311 seconds with return value 0
Press any key to continue . . .
```

60. What are the pros and cons of desktop applications compared to web applications?

⇨ Pros and cons of desktop applications and web applications

- Desktop Applications – Pros

  - ✓ Faster Performance
  - ✓ Can work offline
  - ✓ Better access to system hardware (e.g. printer, camara, files)
  - ✓ More secure for sensitive local data.

- Desktop Applications – Cons

  - ✓ Must be installed on each device.
  - ✓ Platform – dependent (Windows, macOS).
  - ✓ Updates require manual and separate installation.
  - ✓ Not easily accessible from anywhere.

- Web Application – Pros

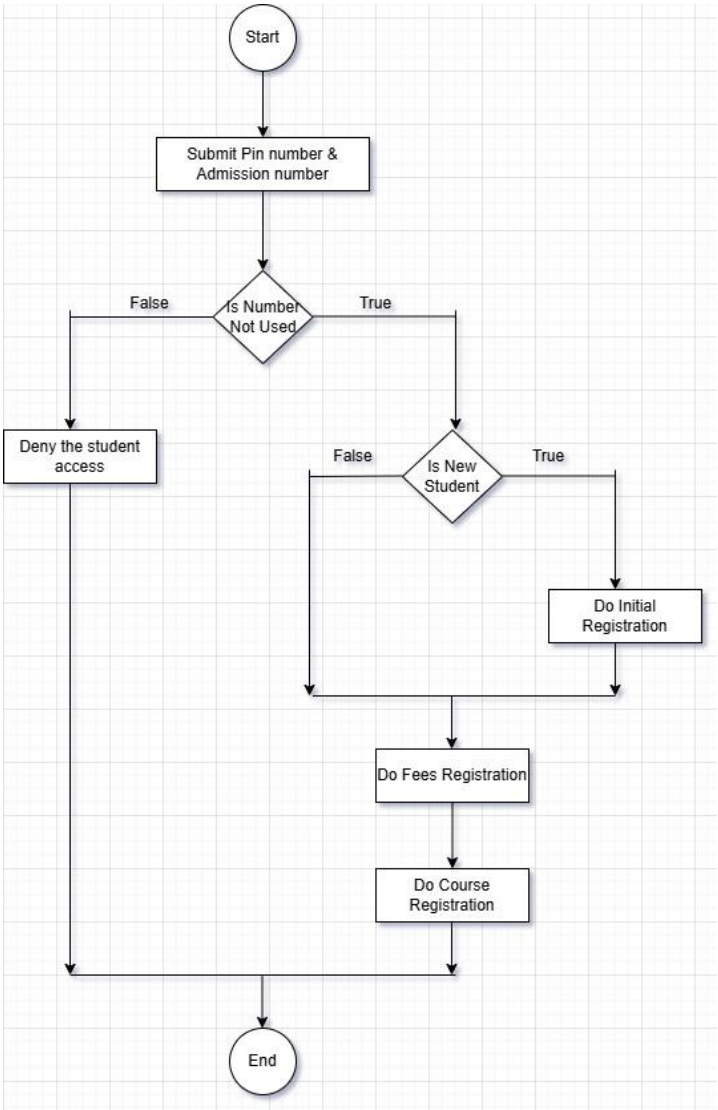
  - ✓ No Installation require
  - ✓ Accessible form anywhere with internet.
  - ✓ Platform-Independent.
  - ✓ Easy to Update.

- Web Application – Cons

  - ✓ Requires internet connection.
  - ✓ Slower compared to desktop apps.
  - ✓ Limited access to device hardware.
  - ✓ Security depends heavily on server and browser protection.

61. Draw a flowchart representing the logic of a basic online registration system.

⇨ Here’s below represent a flowchart :-



62. How do flowcharts help in programming and system design?

⇨ There are key aspects that flowchart help in programming and system design.

- In Programming :-

  - ✓ Planning & Logic
  - ✓ Debugging & Troubleshooting
  - ✓ Modular design
  - ✓ Documentation

- In System Design :-

  - ✓ Process Mapping

- ✓ Communication Bridge
- ✓ Efficiency & Optimization
- ✓ Collaboration